SHREYA BALAKRISHNA

(413) 246-1898 | shreyabalakr@umass.edu | linkedin.com/in/shreya-balakrishna/ | github.com/shrxyo

Education

University of Massachusetts Amherst — Amherst, MA
Master of Science in Computer Science, GPA: 3.789/4 — *Sep. 2024 – May. 2026*
Coursework: Theoretical Machine Learning, Advanced NLP, Information Retrieval, Advanced Algorithms

Vellore Institute of Technology — Tamil Nadu, India
Bachelor of Technology in Computer Science and Engineering, GPA: 8.71/10 — *Sep. 2020 – Jul. 2024*
Coursework: Computer Vision, DBMS, DSA, Computational Social Science, Cloud Computing, Operating Systems

Technical Skills

ML & AI: PyTorch, TensorFlow, RAG, Agentic Systems, LangChain, LangGraph, Supervised Fine-Tuning, RLHF, Reward Modeling, LoRA, vLLM, Vector Databases, OpenAI API, HF Transformers
Data & Analytics: Pandas, NumPy, Scikit-Learn, PySpark, BigQuery, ETL
Frameworks & Tools: FastAPI, Django, REST, WebSocket, Kafka, GCP, AWS, Docker, Kubernetes, CI/CD, Prometheus, Slurm/HPC
Languages: Python, C++, JavaScript, TypeScript, Bash/Shell, R
Certifications: Google Data Analytics Professional (Coursera), AI Analyst (IBM), Data Analytics Using Big Data (AWS, NUS)

Experience

AI Research Extern — Amherst, MA
Oracle — *Jan. 2026 – Present*
- Developed a gradient-free LLM training framework combining **Evolution Strategies** with latent chain-of-thought reasoning, enabling **backpropagation-free** CoT training on **Qwen2.5-3B** with zero intermediate activation storage.
- Designed and benchmarked 3 gradient-free latent architectures (weighted softmax on top-k tokens, residual MLP head, cross-attention head) within the ES training loop.
- Achieved 34.96% test reward on **Weighted Softmax** method on top-10 tokens vs. 24.18% on ES baseline, a **44% relative gain**, outperforming **vanilla COCONUT** (20.00%), validating token-embedding-grounded representations over hidden-state injection.

Graduate Researcher — Amherst, MA
Center for Intelligent Information Retrieval — *Jul. 2025 – Jan. 2026*
- Curated a domain-specific **benchmark** of 500K+ WildChat dialogues filtered by repetition density to study **redundant token generation** in LLM outputs, enabling reproducible measurement of copy behavior across settings.
- Fine-tuned **Llama-3.2-1B-Instruct** to detect and mark redundant content during generation using a custom structured syntax tag, achieving 75.1% train and 71.3% eval accuracy on syntax-compliant generation with a **<4% generalization gap**.
- Validated the reuse-tagging pipeline across **DeepSeek V3, Qwen 3,** and **Kimi K2** under varying repetition conditions, confirming **longest-common-substring** based detection generalizes across model families.

Data Science Intern — Mumbai, India
Metropolis Healthcare — *Jan. 2024 – May 2024*
- Partnered with operations stakeholders to define KPI requirements, then designed and extended **ETL pipelines** processing 2M+ rows from **MySQL** sources with multi-stage transformation and data quality checks.
- Automated **Tableau dashboards** backed by **AWS S3** and monitored pipeline health via **AWS CloudWatch**, eliminating 12+ hours of manual weekly reporting.
- Deployed a production **summarization pipeline**: **T5** for extractive compression and **GPT-3.5** for language generation, with a **RAG layer** grounding outputs in patient-specific clinical records; served via **AI chatbot** with 20% improvement in end-user satisfaction.

Data Science Intern — Singapore, Singapore
National University of Singapore — *Jun. 2023 – Jul. 2023*
- Built SAFESCAN, a **YOLOv8**-based PPE detection system achieving 82% accuracy at **30 FPS** on live video.
- Optimized inference pipeline to maintain **real-time throughput** without sacrificing detection precision; deployed via **Flask** for multi-class monitoring.

Projects

Realtime Chat Platform *(ongoing) | Python, FastAPI, Postgres, pgvector, Redis, Next.js, TypeScript, Docker, Fly.io*
- Building a horizontally-scalable real-time chat platform with AI-assisted threads (RAG over message history), semantic search via pgvector embeddings, and presence/typing indicators over Redis Pub/Sub.
- Targeting sub- 100ms P99 message latency at 10K concurrent WebSocket connections; shipping behind Docker + GitHub Actions CI/CD with structured logging and Prometheus metrics.
- Drove development through a context-engineered agentic AI workflow (cursor, Claude Code): a 300-task plan of scoped agent prompts with phase gated execution to keep output aligned to the architecture.

Code-exploration QA agent | *Python, FastAPI, React, ChromaDB, Llama 3.1 (Groq), SSE*
- Built a full-stack RAG system that clones any GitHub repo, chunks and embeds source files (all-MiniLM-L6-v2) into ChromaDB, and answers natural-language questions grounded in the top-8 retrieved chunks with inline file:line citations.
- Streamed LLM responses token-by-token over Server-Sent Events with multi-turn conversational context, and auto-generated a structured repo-summary card via constrained LLM-to-JSON output.
- Designed a provider-abstraction layer that hot-swaps local (Ollama) and cloud (Groq) LLM backends via a single env var, then deployed backend and Reactfrontend on Vercel with live /health checks.